

Estimating critical stream power for bedload transport calculations in gravel-bed rivers

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Abstract

The physical basis of Bagnold's equation for critical stream power is examined. Although the general approach is well founded, the original equation can be criticised for (1) failing to distinguish between the grain size in transport and the grain size that represents bed roughness; (2) requiring a knowledge of critical flow depth in addition to gross channel properties; (3) failing to allow for hiding and protrusion effects on the mobility of mixed-size stream beds. More general equations are presented which overcome these limitations. They require a knowledge of channel slope, but not depth. The analysis suggests that the presence of form resistance in addition to grain roughness does not affect the calculation of critical power. Critical power is expected to depend more sensitively on grain size between reaches than within reaches.

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1. Introduction

Bedload transport in gravel-bed rivers is commonly assumed to occur at a very low rate up to a certain critical level of streamflow, then to increase at a faster-than-linear rate with flow above this threshold. The flow variable used to predict bedload transport is usually shear stress, as for example in the well known equation of Meyer-Peter and Müller (1948). In uniform flow the mean shear stress, averaged across the channel, depends on the depth-slope product:

$$\tau = \rho g d S \quad (1)$$

where ρ denotes the density of water, g the acceleration due to gravity, d the mean flow depth, and S the bed and water-surface gradient. Bagnold (1977, 1980) proposed that bedload transport could alternatively be predicted from the mean value of stream power per unit bed area, defined as

$$\omega = \rho g Q S / w = \tau U \quad (2)$$

where Q denotes the water discharge, w the width of the river, and U the mean velocity. It should be noted that Bagnold omitted g in his equations; this has led to much confusion in the subsequent literature. Eq. (2) quantifies the rate of loss of potential energy as water flows downhill, and thus the power potentially available for performing geomorphic work. Bagnold

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proposed that bedload transport rate increases non-linearly with stream power above a threshold or critical value which I shall denote by ω_c .

Bagnold's approach is conceptually attractive insofar as it treats rivers as transporting (and therefore work-performing) machines with explicit attention to their efficiency. It is also pragmatically convenient in that stream power can be calculated from gross channel properties (width and slope), together with the discharge provided by the catchment, without needing to know within-channel flow properties such as depth or velocity. Discharge is essentially constant from one junction to the next along a river and may be known from hydrometric records or predictable from a hydrological model, whereas depth (needed for the calculation of shear stress) is locally variable and not routinely measured. Partly for these reasons, and also because it has performed well in some comparative tests (Gomez and Church, 1989; Martin and Church, 2000), Bagnold's transport equation has been used quite widely in the geomorphology literature.

There has been some re-evaluation of the efficiency factor in Bagnold's transport equation (e.g. Bagnold, 1980; Martin and Church, 2000), but all users of any version of the equation appear to have accepted Bagnold's (1980) proposal for specifying the critical power in terms of river properties:

$$\omega_c = c_1 D^{1.5} \log(c_2 d/D) \quad (3)$$

where c_1 and c_2 are numerical constants, D denotes the diameter of mobilised particles, and the logarithm is to base 10. Bagnold (1980) gave $c_1=290$ and $c_2=12$ without fully explaining his derivation. These values are for D in the same units as d , for example metres; and, as noted below, the predicted values of ω_c are not in the standard units of W m^{-2} .

The lack of attention to the specification of the threshold is surprising considering that all gravel transport equations are extremely sensitive to the threshold value of whichever flow variable is used, and that there is an extensive literature on threshold shear stresses for poorly sorted river beds (see for example Andrews, 1983; Komar, 1987; Ashworth and Ferguson, 1989). Petit et al. (2005) take an important first step towards assessing whether Eq. (3) is universally valid. They present a useful and interesting synthesis of empirically derived critical specific stream power values for streams in Belgium, and

compare them with a version of Eq. (3). Their results are based on the movement of tracer pebbles in 14 streams and rivers with slopes ranging from <0.002 to 0.05 . The inferred values of critical power show clear correlations with D within each data set, as expected from Eq. (3), but with displacements between data sets. Petit et al. (2005) speculate that these displacements reflect the varying importance of bedform resistance to flow in small, medium, and large rivers. Critical stream power was found to be higher in channel styles where a higher proportion of the total shear stress (as given by Eq. (1)) is dissipated in overcoming bedform resistance.

This note was inspired by reading the work of Petit et al. (2005) but is largely complementary to it. I consider the question of critical stream power from a theoretical point of view. I show that Bagnold's expression for critical power can be criticised on several grounds, and present alternative equations based on a more detailed analysis which overcomes these problems. I then discuss the implications for differences in critical power within and between reaches. I argue on physical grounds that critical power is not affected by form resistance, but is higher in reaches with coarser beds and also varies somewhat with channel slope; this provides an alternative explanation for the pattern of differences found by Petit et al. (2005). A detailed comparison with their data is not possible at this time for lack of sufficient published detail about the characteristics of their individual reaches. The theoretical rate of increase of critical power with grain size within reaches is less than suggested by the empirical curves presented by Petit et al. and previous workers, but the discrepancy is mainly due to their aggregation of results from different reaches.

2. Derivation of general equations for critical stream power

Bagnold (1980) derived Eq. (3) from the identity $\omega_c = \tau_c U_c$ by relating τ_c to D through Shields' criterion for incipient motion, and relating U_c to d/D using a flow resistance equation of logarithmic type. He gives few further details of the derivation but it is fairly easy to reconstruct. Before doing so, though, a first criticism of Bagnold's equation is

immediately evident. The symbol D is used for grain diameter in two different contexts: the size entrained by the flow, and the size that represents flow resistance. These sizes are not normally the same. Flow resistance is dominated by the coarser grains in the bed, whereas transport in all but the biggest floods is preferentially of the finer grains, so it is inappropriate to use the same diameter for both. I therefore distinguish between the size entrained by the flow, denoted by D_i , and a representative diameter (for example the median or geometric mean) for the bed surface as a whole, denoted by D_b .

2.1. Derivation using separate grain sizes

The logarithmic flow resistance equation for open channels, first proposed by [Keulegan \(1938\)](#), takes the general form

$$U/u^* = A \ln(Bd/k_s) = 2.30A \log(Bd/k_s) \quad (4)$$

where $u^* = (\tau/\rho)^{0.5}$ is the shear velocity of the flow and k_s is the roughness height of the bed. Eq. (4) is based on integrating the ‘law of the wall’ for turbulent boundary layers:

$$u/u^* = (1/\kappa) \ln(z/z_0) = (1/\kappa) \ln(30z/k_s) \quad (5)$$

where u is velocity at height z , κ is von Karman’s constant, and z_0 is the height at which u goes to zero. The most widely accepted experimental value of von Karman’s constant is 0.417, but it was originally regarded as ≈ 0.4 which gives $A=2.03$ or $2.30A=5.75$ in Eq. (4). If Eq. (5) holds throughout the flow depth, the mean velocity U is found by integration to be the same as u at height d/e ($\approx 0.37d$) where e is the root of natural logarithms; this gives $B=11.0$ in Eq. (4). [Bagnold \(1980\)](#) took the mean velocity to be at $0.4d$ which gives $B=12$, close to [Keulegan’s \(1938\)](#) value of 12.2.

The roughness height for a well-sorted rough bed with no bedforms is expected to equate with D_b , but in most gravel-bed rivers flow resistance is dominated by small-scale form drag on large protruding clasts or pebble clusters, and possibly form drag associated with larger bedforms, bank protrusions, or large woody debris. The value of k_s in Eqs. (4) and (5) also has to subsume any channel shape factor that arises in integrating the log law across the channel

(see [Yen, 2002](#)). Empirical analyses by [Hey \(1979\)](#) and many subsequent workers have suggested that k_s scales as a small multiple of D_{84} or D_{90} , and thus a larger multiple of D_{50} . For generality one can write

$$K_s = mD_b \quad (6)$$

where m is a parameter of the particular channel bed of interest and can have a value from ~ 1 to >10 depending on how poorly sorted the bed is and the degree of form (rather than grain) roughness in the channel. The grain resistance of a well-sorted bed is represented by $m=1$. A low value of what [Petit et al. \(2005\)](#) denote by τ'/τ , the ratio of effective to total shear stress, would correspond to a high value of m .

By substituting the logarithmic resistance law (4), with k_s given by (6), in the identity $\omega_{ci} = \tau_{ci} U_{ci}$ an expression is obtained for the critical stream power to move size D_i :

$$\omega_{ci} = \frac{2.30\tau_{ci}^{1.5}}{\kappa\rho^{0.5}} \log\left(\frac{30d_{ci}}{emD_b}\right) \quad (7)$$

The critical shear stress τ_{ci} can now be related to the mobilised grain size using Shields’ criterion:

$$\tau_{ci} = \theta_{ci}(\rho_s - \rho)gD_i = \theta_{ci}\rho RgD_i \quad (8)$$

where θ_{ci} denotes the dimensionless or Shields stress for entrainment of size D_i and ρ_s and R denote sediment density and submerged specific gravity. Substituting Eq. (8) in Eq. (7) yields

$$\omega_{ci} = \frac{2.30\rho(\theta_{ci}RgD_i)^{1.5}}{\kappa} \log\left(\frac{30d_{ci}}{emD_b}\right) \quad (9)$$

It ought now to be possible to recover Bagnold’s value $c_1=290$ in Eq. (3) by using $\theta_{ci}=0.04$, $R=1.6$, $\rho=1000 \text{ kg m}^{-3}$, $g=9.8 \text{ m s}^{-2}$, and $\kappa=0.4$ in Eq. (9). However, these values give $c_1 \approx 2860$, which is higher by a factor of just over 9.8. As already noted, Bagnold omitted g from the definitional equation of stream power and it seems likely that he did so again here. [Gomez and Church \(1989\)](#) and [Martin and Church \(2000\)](#) similarly quote an equation for ω_c that is equivalent to Eq. (9) here except that it contains $g^{0.5}$ instead of $g^{1.5}$. Their version has dimension $\text{M L}^{-1} \text{ T}^{-1}$, which is commensurate with bedload transport rate, whereas Eq. (9) has dimension M T^{-3} which is correct for power per unit area; in SI units, ω_{ci} from Eq. (9) is measured in W m^{-2} . Using more precise

values of g and κ , and the conventional value of 1.65 for R , does not alter c_1 appreciably. Variation in θ_{ci} because of hiding and protrusion effects is considered later, as is the effect of form resistance.

2.2. Alternative resistance formulation

An obvious alternative to the logarithmic flow resistance law (Eq. (4)) is the Manning–Strickler equation in which Manning's n is related to the 1/6 power of D_b when resistance is entirely due to grain roughness. This can be written in the non-dimensional form

$$U/u^* = a(d/D_b)^{1/6} \quad (10)$$

where a is a constant and D_b is normally defined as the median surface grain diameter. The Strickler relation is quoted in many different forms in the literature, partly depending on the units assumed for D_b , but all forms I have found correspond to a value of either 6.7 or 8.2 for a in Eq. (10). The equivalent of Eq. (9) is now

$$\omega_{ci} = a\rho(\theta_{ci}RgD_i)^{3/2} \left(\frac{d_{ci}}{D_b} \right)^{1/6}. \quad (11)$$

Eqs. (9) and (11) are similar in all but the functional form of the dependence on the relative submergence d_{ci}/D_b . As will be seen later, for typical values of this ratio the two equations predict approximately the same critical stream power.

2.3. Eliminating the dependence on depth

A second obvious disadvantage of Bagnold's equation for critical stream power, when expressed in the form of Eq. (3) or (9), is that it requires a knowledge of flow depth. This criticism applies equally to Eq. (11). It removes what was identified earlier as the main advantage of using stream power rather than shear stress to predict bedload transport: that local flow properties do not need to be known. Moreover, application of any of Eqs. (3), (9), or (11) requires not just a flow depth at bankfull or other fixed discharge, but a knowledge of the flow depth at each of the discharges at which different grain sizes start to move. It is not clear whether previous authors have realised this. Bagnold simply defined d in Eq. (3) as

depth, whereas the derivation makes it clear that it is a critical depth, and therefore must be different for different grain sizes if entrainment is size selective and occurs at progressively higher stages for coarser grains. This appears to be a significant disadvantage of the way Bagnold specified the threshold, and the lack of clarity about it in his papers may have led to incorrect calculations in some subsequent literature (in the same way that his omission of g from all equations led to order-of-magnitude plotting errors in graphs in several papers in the 1980s).

Fortunately it is easy to eliminate critical depth from Eqs. (9) and (11) and recast them in a form which uses only readily available channel properties. All that is required is to invert Eq. (1) and then use Eq. (8) to obtain

$$d_{ci} = \frac{\tau_{ci}}{\rho g S} = \frac{\theta_{ci} R D_i}{S}. \quad (12)$$

Eq. (9), based on logarithmic flow resistance, now becomes

$$\omega_{ci} = \frac{2.30}{\kappa} \rho (\theta_{ci} R g D_i)^{1.5} \log \left(\frac{30 \theta_{ci} R}{em S} \frac{D_i}{D_b} \right). \quad (13)$$

Similarly, if flow resistance is represented by the Manning–Strickler law, Eq. (11) becomes

$$\begin{aligned} \omega_{ci} &= a\rho(\theta_{ci}RgD_i)^{3/2} \left(\frac{\theta_{ci}RD_i}{SD_b} \right)^{1/6} \\ &= \frac{a\rho g^{3/2} (\theta_{ci}RD_i)^{5/3}}{(SD_b)^{1/6}}. \end{aligned} \quad (14)$$

An important point to note about Eqs. (13) and (14) is that, when different rivers or reaches are compared, the critical stream power for a given absolute grain size D_i depends on the channel gradient, S , as well as the representative bed diameter, D_b . This is directly analogous to the way in which depth affects critical power in Bagnold's equation; all that has been done is to replace depth by one of its main controls, channel gradient. The magnitude of this effect is considered later.

2.4. Relative-size effects

The third main criticism that can be levelled at Bagnold's equation for critical stream power is that it

makes no allowance for the effects of grain hiding and protrusion on the mobility of different sizes within a poorly sorted gravel bed. A constant value of c_1 in Eq. (3) is possible only if θ_{ci} in Eq. (9) also takes a constant value, for example 0.04 as assumed by Bagnold. In reality θ_{ci} depends on the relative grain size D_i/D_b : smaller-than-average grains tend to be hidden by their larger neighbours and therefore have higher θ_{ci} , whereas bigger-than-average grains tend to protrude and to have lower θ_{ci} . These effects had been discussed in some of the technical literature before Bagnold published his work on bedload transport (e.g. Einstein, 1950; Egiazaroff, 1965; Fenton and Abbott, 1977), but only became widely known to geomorphologists in the 1980s following the work of Parker et al. (1982) and Andrews (1983). Several alternative hiding functions to quantify these effects have been proposed. The addition of any such function to Bagnold's equation for critical power should improve its performance, in the same way that Ferguson (1994) showed that inclusion of a hiding function in previous equations for critical discharge gave a better fit to empirical data. The simplest and most widely used hiding function is the power law proposed by Andrews (1983). In present notation this can be written as:

$$\frac{\theta_{ci}}{\theta_{cb}} = \left(\frac{D_i}{D_b}\right)^{-b} \quad (15)$$

where b is a hiding factor (not to be confused with B in Eq. (4) for flow resistance) which takes a value between 0 and 1. The limit $b=0$ corresponds to no hiding and protrusion, as implicit in Bagnold's expression for critical power. The limit $b=1$ is reached when hiding and protrusion effects completely offset the intrinsically lower mobility of coarser particles so that the critical shear stress (in N m^{-2}) is the same for all D_i . This is one definition of the condition of perfect 'equal mobility' which has been much discussed since Parker et al. (1982) and Andrews (1983) first proposed that natural rivers are close to this state. A wealth of empirical evidence from both field and flume has confirmed that b is closer to 1 than 0, but the precise value depends on the method used to estimate it. Estimates from the fractional transport rates of different grain sizes typically give $b \approx 0.7$ at low transport rates, increas-

ing towards $b=1$ in fully mobile conditions in big floods or high flows in the flume (e.g. Ashworth and Ferguson, 1989; Wilcock and Southard, 1988). Estimates from the biggest grain size moving in different flows tend to give rather lower values of b , typically 0.5 to 0.7 (e.g. Komar, 1987; Ashworth and Ferguson, 1989). Since Petit et al. (2005) obtained their data from tracer-pebble experiments rather than bulk transport measurements, $b \approx 0.6$ with an uncertainty of at least ± 0.1 seems appropriate for comparison with their results.

Substitution of Eq. (15) in Eqs. (13) and (14) gives what should be more accurate critical-power equations than Eq. (9) or (11):

$$\omega_{ci} = \frac{2.30}{\kappa} \rho (\theta_{cb} R g D_b)^{1.5} \log \left[\frac{30 \theta_{cb} R}{em S} \left(\frac{D_i}{D_b}\right)^{1-b} \right] \times \left(\frac{D_i}{D_b}\right)^{1.5(1-b)} \quad (16)$$

using a logarithmic flow resistance law as adopted by Bagnold, or

$$\omega_{ci} = a \rho (\theta_{cb} R g D_b)^{3/2} \left(\frac{\theta_{cb} R}{S}\right)^{1/6} \left(\frac{D_i}{D_b}\right)^{5(1-b)/3} \quad (17)$$

using the Manning–Strickler relation. These equations are proposed as the most general theoretically based equations for the critical specific stream power to move grains of any size in a mixed gravel bed.

Practical use of these equations requires setting values for the various hydraulic constants and parameters. With $b=0.6 \pm 0.1$, Eq. (17) implies that ω_{ci} increases as the 0.50 to 0.83 power of D_i . The dependence of ω_{ci} on D_i in Eq. (16) is not an exact power law, but with plausible values of the parameters and channel variables it is not perceptibly nonlinear in a log–log plot. For most situations it is reasonable to assume $\kappa=0.4$, $\rho=1000 \text{ kg m}^{-3}$, $g=9.8 \text{ m s}^{-2}$, $R=1.65$, and $\theta_{cb}=0.045$ (though the last assumption is reviewed later). If the hiding factor is set at $b=0.6$, the equations simplify to

$$\omega_{ci} = 0.113 D_b^{1.5} \log \left[\frac{0.73}{S} \left(\frac{D_i}{d_b}\right)^{0.4} \right] \left(\frac{D_i}{d_b}\right)^{0.6} \quad (18)$$

using the logarithmic resistance law with $m=1$, or

$$\omega_{ci} = 0.104 \frac{D_b^{1.5}}{S^{0.17}} \left(\frac{D_i}{D_b} \right)^{0.67} \quad (19)$$

using the Manning–Strickler resistance law with $a=8.2$. The coefficients in these two equations have been altered to yield ω_{ci} in W m^{-2} when grain sizes are given in mm, rather than m as previously.

3. Variation of critical power within and between reaches

Predicted patterns of within-reach and between-reach variation in critical stream power according to Eqs. (18) and (19) are illustrated in Fig. 1. The within-reach variation is shown for grain sizes from half to twice the local bed mean and the between-reach variation is shown for combinations of two channel gradients (0.002 and 0.02) and two median bed diameters (30 mm and 100 mm). The two equations give almost identical predictions for the gentler slope, and agree to within about 10% at the steeper slope, whereas both equations predict big differences between reaches according to either bed grain size or channel slope.

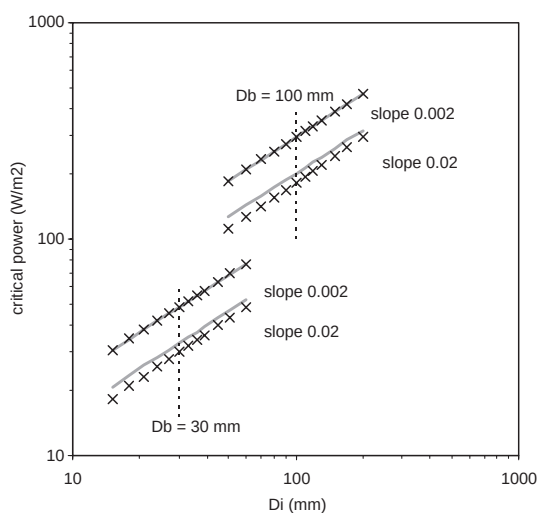


Fig. 1. Within-reach variation in critical stream power as predicted by Eqs. (15) (using logarithmic flow resistance; plotted as crosses) and (16) (Manning–Strickler flow resistance; plotted as lines). Predictions are shown for four hypothetical reaches with slope 0.002 or 0.02 and median bed grain size (D_b) 30 mm or 100 mm.

Eq. (19) implies a 32% reduction in critical power with a tenfold increase in channel slope, and Fig. 1 shows that Eq. (18) is at least as sensitive to this controlling variable. The sensitivity to median bed grain size (D_b) is greater still, though the exact effect depends on the hiding factor. With $b=0.6$, it can be deduced from Eq. (19) that doubling D_b with all else fixed causes a 78% increase in the critical power to move a given grain size. The four pairs of lines in Fig. 1 are consequently well separated. The rivers for which results are collated by Petit et al. (2005) vary greatly in slope (0.001 to 0.07) and D_b (15–230 mm), so they too would be expected to plot as a series of offset lines rather than along a single trend. Fig. 2 in their paper shows such a pattern. The trendlines for individual rivers are, however, quite variable in slope, and mostly rather steeper than the theoretical curves in my Fig. 1. One possible explanation for the discrepancy would be that the hiding factor differs between rivers and is generally lower than 0.6, but there is no obvious rationale for this. Another factor which could be relevant is that the trendlines for several of the Belgian rivers are fitted to composite sets of data, combining results of tracer experiments in several different reaches with different values of D_b . In some cases there is a three- or four-fold variation in D_b (Petit et al., 2005, Table 2). As can be seen in Fig. 1, merging data from reaches with the same slope but different D_b would give a composite curve which is steeper than the individual within-reach curves.

The effect of reach conditions on critical power can be isolated by setting $D_i=D_b$ in any of Eqs. (16)–(19) and thus eliminating the effects of hiding and protrusion within reaches. This leads to a power-law dependence of ω_{cb} on D_b with an exponent of 1.5, and a far less sensitive inverse dependence on slope. Since coarser beds tend to be associated with steeper slopes, the overall relation between ω_{cb} and D_b within a region would be expected to have an exponent somewhat below 1.5. This prediction agrees quite well with the powers of 1.30, 1.69, and 1.44 fitted by Williams (1983), Costa (1983), and Petit et al. (2005) for compilations of field data.

3.1. Effect of form resistance

The numerical calculations thus far in this paper have been based on resistance from grain roughness

alone: $m=1$ in the logarithmic flow resistance law, or $a=8.2$ in the dimensionless Manning–Strickler relation. The general forms of the equations (Eqs. (7) and (11)) provide a physical basis for considering the hypothesis of Petit et al. (2005) that channels with form as well as skin resistance will have higher critical stream power, all other things being equal.

As noted earlier the actual stream power ω depends only on the gross properties of the channel as a container for the flow (i.e. channel width and bed gradient), and not on the local flow properties (depth and velocity). More specifically, ω is unaffected by the existence and amount of form resistance to flow. This can be seen by writing it as $\omega = \rho g Q S / w = \rho g d U S$ where d is mean depth and U is section-average velocity. Form resistance leads to an increase in d and a corresponding reduction in U , with the unit discharge ($Q/w = dU$) unchanged.

In contrast, shear stress is normally estimated as $\tau = \rho g d S$, so form resistance leads to the same proportional increase in τ as in d . Entrainment and bedload transport are properly estimated from only the grain-roughness component of shear stress, τ' say. It is convenient to regard this component as corresponding to an effective depth (d') that would exist in the absence of form resistance: $\tau' = \rho g d' S$ (Einstein and Barbarossa, 1952). It will also be useful to write U' for the mean velocity that would be found in the absence of form resistance. So $d' U' = d U$; and if d exceeds d' by a factor f (>0) as a result of form resistance, so that $d = d'(1+f)$, then $\tau = \tau'(1+f)$ and $U = U'/(1+f)$. This implies an increase in Manning's n over the value (n' say) corresponding to grain roughness alone (Yen, 2002). The magnitude of this increase is given by $n = n'(1+f)^{5/3}$ if the gradient S remains the same, and can be considerable. Petit et al. (2005) report values of τ'/τ ranging from 0.7 to <0.3 , which corresponds to f from 0.4 to >2 and implies values of n anywhere from <2 to >10 times what would be expected for grain roughness alone using the Strickler relation.

Although stream power is clearly determined by external variables and unaffected by the presence or amount of form resistance, it is less obvious whether form resistance has implications for the critical stream power ω_{ci} for movement of grains of diameter D_i . Petit et al. (2005) assume that ω_{ci} increases with form resistance, on the basis that more of the stream power

is expended in overcoming resistance and less is available for sediment transport, and suggest that this is the explanation for between-river differences in ω_{ci} . However, despite the apparent plausibility of this argument, the theoretical analysis presented above can be extended to show that critical stream power is in fact unaffected by form resistance.

This can be seen by reviewing carefully the derivation of ω_c as the product of τ_c and U_c . The derivation related τ_c to D using Shields' criterion, then represented the corresponding value of U_c as the product of u^* and a function of d/D . The shear velocity u^* was defined as $(\tau/\rho)^{1/2}$ in which τ can again be substituted using Shields. In the presence of significant form resistance, it is clear that Shields' criterion should be used to define a critical grain shear stress (τ'_{ci}) and that the total shear stress at the onset of motion will be higher: $\tau_{ci} = (1+f)\tau'_{ci}$. The critical depth will be higher by the same factor. The corresponding velocity to use in $\omega_c = \tau_c U_c$ should be calculated on the basis of this increased depth and also using a higher total flow resistance (higher k_s or n , with the implication of higher m or lower a in Eqs. (6) or (10)). Using the basic form of Manning's equation ($U = d^{2/3} S^{1/2} n^{-1}$), assuming constant slope, and continuing to use primes for values in the absence of form resistance, it can be seen that $U_c/U'_c = (d/d')^{2/3} (n/n')^{-1} = (1+f)^{2/3} (1+f)^{-5/3} = (1+f)^{-1}$. Thus $\omega_c = \tau_c U_c$ is invariant: the increase in τ_c is balanced by the reduction in U_c . Although this result is not what would intuitively be expected, it is very convenient: it means that critical stream power, like actual stream power, depends only on gross channel properties and not on the details of within-channel flow, even in the presence of form resistance.

If this analysis is correct, it implies that the tendency identified by Petit et al. (2005) for smaller headwater streams to have higher critical power for a given grain size cannot be explained by higher form resistance. Between-stream differences in critical power should instead be explicable in terms of differences in bed grain size and channel slope. If headwater streams in general have coarser beds than larger channels, both Bagnold's equation and my extended version of his analysis point to higher critical power, though offset to some extent by lower depth (in Bagnold's equation) or higher slope (in my

equations). A thorough test of this is not possible, because the tables in [Petit et al. \(2005\)](#) do not provide details of slope and bed grain size for individual reaches. It should also be noted that critical power could differ between sites because of differences in the critical Shields stress (θ_{cb}) for the average bed grain size. [Bagnold \(1980\)](#) assumed a value of 0.04. [Fig. 1](#) is based on $\theta_{cb}=0.045$, which [Buffington and Montgomery \(1997\)](#) propose as a good representative value for a visual or competence-based threshold. But there is evidence to suggest that θ_{cb} is higher than this in steep streams ([Hassan and Church, 2000](#); [Shvidchenko and Pender, 2000](#)), either because of the development of surface structure or because of the effect of low relative depth (d/D) on flow resistance. A higher value of θ_{cb} would increase critical stream power relative to the value given by Eq. (18) or (19).

4. Conclusions

[Bagnold's \(1980\)](#) equation for critical stream power has caused much confusion because of his omission of the gravity acceleration from many equations and his failure to make this clear. The equation has a logical physical basis, but the form which he presented (and subsequent researchers have used) can be criticised on three grounds. First, no distinction is made between the grain size transported by the flow and the (usually higher) grain size which characterises the bed roughness. Second, the formula requires a knowledge of flow depth; this depth should be the critical depth for motion of each grain size, not a constant value such as the bankfull depth. Third, [Bagnold](#) assumed the same critical Shields stress applies to all grain sizes, ignoring hiding and protrusion effects which are now known to be strong.

This paper has presented a more detailed derivation on the same lines as [Bagnold's](#) but avoiding these deficiencies. The depth term in [Bagnold's](#) equation is replaced by a channel slope term, and a relative grain-size term is added. Two versions are given, one using (like [Bagnold](#)) a logarithmic flow resistance law and the other using the Manning–Strickler law to obtain a slightly simpler result. The equations can be used to estimate the critical power to move the average bed grain size, or grains coarser or finer than this, from a

knowledge only of channel gradient and average bed grain size. Empirical testing of these theoretically based equations is evidently desirable.

Contrary to intuition, and to the hypothesis of [Petit et al. \(2005\)](#), the analysis suggests that the critical power for a given channel does not depend on the proportion of total shear stress that is related to form resistance rather than grain roughness. This points to a significant advantage of [Bagnold's](#) approach to bedload transport: excess stream power depends only on the externally imposed discharge and the gross channel properties of slope and width, without requiring any knowledge of within-channel flow properties such as depth, velocity, or resistance. Assumptions must, however, be made about the critical Shields stress to move the average grain size and the parameter that quantifies hiding and protrusion effects.

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